

Using GPT-3 to Create General Purpose Assistance Model for MIT World Peace University

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Abstract— The MIT-GPT project is a college query assistant built using the OpenAI API and LangChain, which aims to provide accurate and relevant responses to MIT-WPU-related queries. With the increasing demand for online education and the need for instant information, having an AI-powered assistant to answer college-related queries can significantly benefit students, faculty, and staff alike. This technical report provides an overview of the MIT-GPT project, including its architecture, data sources, and performance evaluation.

Keywords— Generative Pre-Training, OpenAI, test-davinci-003, MIT World Peace University

I. INTRODUCTION

Natural language processing [1] has experienced enormous growth in recent years, with several cutting-edge technologies being created to allow robots to comprehend and produce human language. The Generative Transformer 3 (GPT-3) model [2] which has been hailed as one of the most potent language models ever made, is one example of such a technique. GPT-3 model has several uses in a variety of industries, including banking, healthcare, and education. GPT-3 can be used to create models that, among other things, generate exam questions and offer feedback to students in the context of education. In this study, we investigate the application of a GPT-3-based model in the context of higher education. We look at the model's potential to assist students. GPT-3 can be used to create models that, among other things, answer questions related to college classes, locations of offices and other general questions. In this study, we investigate the application of a GPT-3-based model in the context of higher education. We investigate how the model may produce study materials and offer immediate feedback on assignments to aid students in learning more efficiently. We start by giving a general summary of the GPT-3 model's features. The process for creating the GPT-3 model for use in colleges is then discussed, along with the data sources that were used to construct and evaluate the model. We also give the outcomes of our tests, which show how well the model creates study materials and offers immediate feedback. Finally, we go over the consequences of our findings and how they might affect the teaching profession. We contend that applying GPT-3-based

models to college instruction can greatly improve students' learning opportunities and aid them in more successfully achieving their academic objectives.

II. ARCHITECTURE

As shown in Fig. 1, the MIT-GPT project uses a client-server architecture. The client is a web-based interface allowing users to input queries, and the server is the system that processes the queries using OpenAI API[3] and LangChain. A robust set of natural language processing tools and algorithms are available through the OpenAI API that can be used to comprehend and produce human language. The processing system in this architecture is best suited for the programming language LangChain, which was created expressly for creating systems for natural language processing. Clients send user queries to the server, which then processes them using the OpenAI API and LangChain and sends the answers back to the client. The outcome is text-based responses, like answers to queries related to MIT World Peace University. web-based interface provides a user-friendly experience for inputting queries and receiving responses. The system is responsible for processing the queries, sending them to the OpenAI API for language processing, and then using LangChain to and the responses based on the college-specific dataset.

III. DATA SOURCES

The dataset used for training and reining the MIT-GPT model is the college-specific dataset obtained from our college. This study paper's college-specific dataset includes a wide range of data that is pertinent to college students, teachers, and administrators. The dataset serves as a complete resource for the modification of the GPT-3 model because it is created to include a wide range of topics and information relevant to the collegiate domain. The dataset contains details on several courses the college offers, such as course descriptions, prerequisites, and requirements. It also includes details on the faculty members, such as their specialties, areas of interest in Research and contact information. The dataset also contains information on courses and professors as well as specifics on admission requirements, such as details on the application

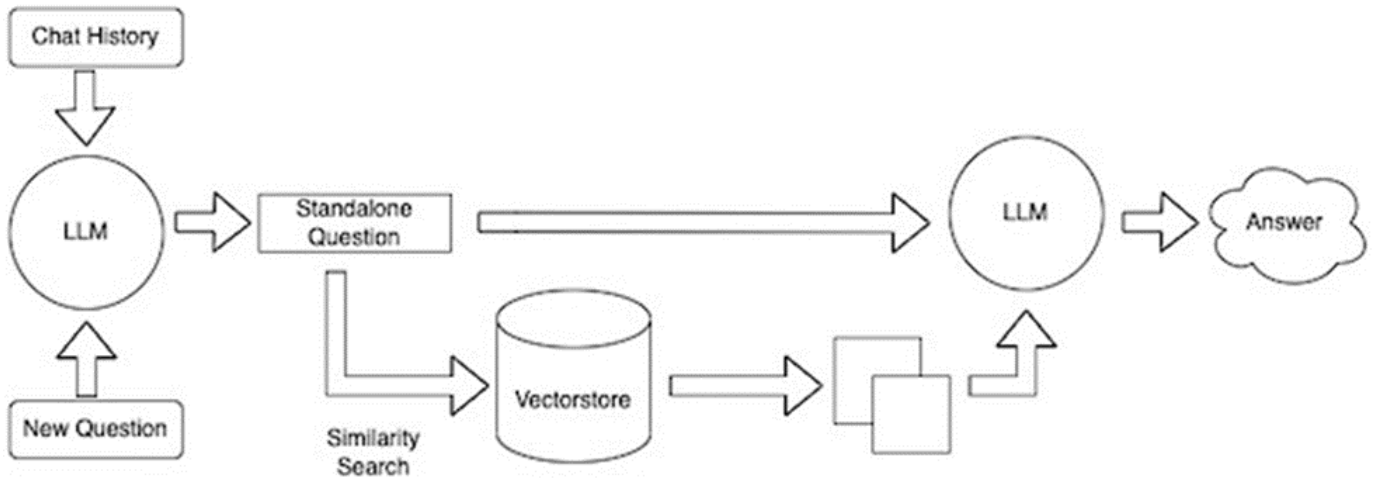


Fig. 1 - Architecture of MIT-GPT

procedure, deadlines, and needed paperwork.

It also includes specifics about the exam procedure, such as the exam timetable, format, and grading guidelines.

The dataset also contains facts about the dining halls, athletic facilities, student centers, library, computer laboratories, and student centers on campus. It is a helpful tool for students and guests because it gives a thorough picture of the college campus and its amenities.

The dataset is used to fine-tune the OpenAI GPT-3.5 language model using the LangChain tool, allowing for domain-specific customization.

A. Model Training and Customization:

The model used in the MIT-GPT project is based on OpenAI's GPT-3.5 architecture, which is a state-of-the-art language model that has been trained on a large corpus of text data. Text-Davinci-003[4] is the model that has been utilized in creation of MIT-GPT. This recent addition to the InstructGPT models solves many backdrops that existed. capacity of Text-Davinci-003 to execute few-shot learning, which enables the model to swiftly adapt to new tasks with only a few samples of training data, is one of its important characteristics. As a result, it is incredibly flexible and versatile and can carry out a variety of language-related tasks.

To customize the model for college-related queries, the LangChain tool is used to fine-tune the model using our college-specific dataset. The model is trained on the specific language and context used in the college domain using examples and prompts from the college dataset that the researcher can supply using the LangChain tool.

By LangChain, the GPT-3 model is enhanced to comprehend the particular jargon, phrases, and idioms in the college domain, making it more suitable for producing pertinent and precise answers to queries pertaining to colleges. The model gains an understanding of college students' tastes and wants thanks to this modification, which enables it to produce responses that are more beneficial to specific users. This fine-tuning process helps the model understand and generate responses specific to

the college domain, making the MIT-GPT assistant highly accurate and relevant for college-related queries.

B. Performance Evaluation

The performance of the MIT-GPT assistant is evaluated based on several metrics, including accuracy, relevance, and response time. Accuracy is measured by comparing the assistant's responses with the ground truth information obtained from the college dataset. Relevance is assessed by evaluating the appropriateness and usefulness of the assistant's responses in the context of college-related queries. Response time is measured as the time the assistant takes to generate responses, and efforts are made to optimize the system for faster response times.

C. Figure Captions

Preliminary results from the MIT-GPT project have shown promising performance in terms of accuracy and relevance. The assistant can generate accurate responses to college-specific questions as shown in Fig. 2 and 3.

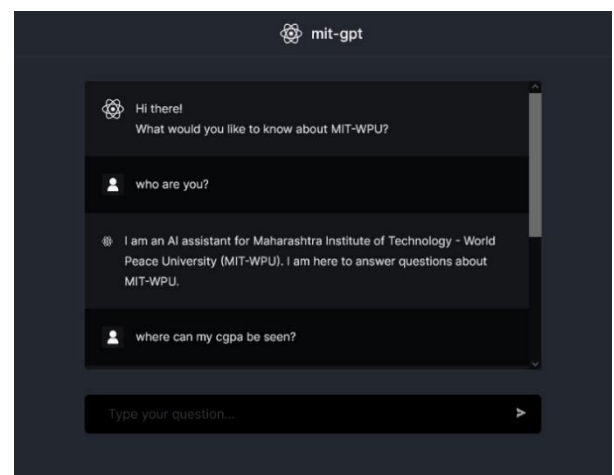


Fig. 1 Example of Output

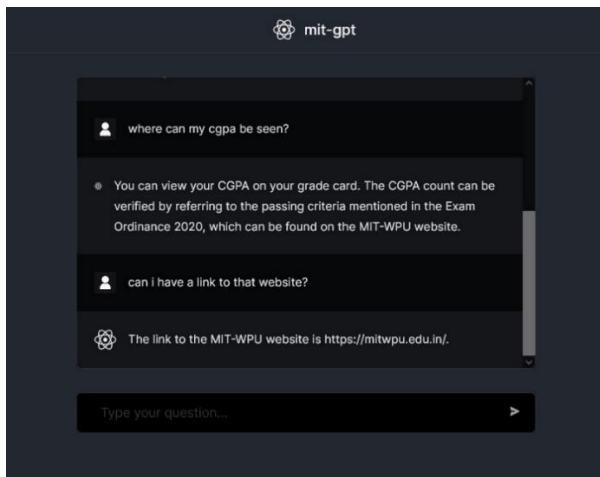


Fig. 3 Example of Output

D. Security and Privacy

Security and privacy are paramount as the MIT-GPT assistant deals with sensitive college-related information. The system is designed to handle data in a secure way to ensure that sensitive information is protected from unauthorized access, modification, or disclosure. To achieve this, the system implements encryption protocols to protect both data transmission and storage. In order to guarantee that only individuals with permission can access sensitive data, the system additionally uses access controls and authentication procedures. According to a user's job or level of authorization, access controls involve restricting access to particular parts of the system or data. A user with administrative rights, for instance, might have access to more information than a regular user. Before allowing access to the system, authentication procedures are employed to confirm a user's identity. Techniques like using students specific Student ID for login along with stated passwords, these could also connect to their direct login to the MIT portal. To prevent data loss in the event of system failures or natural disasters, the system also uses frequent backups and disaster recovery protocols. While disaster recovery processes involve developing a strategy to recover data and restore operations in the case of a disaster like a fire, flood, or attack, frequent backups make sure that data is regularly saved and stored in a secure location. is managed securely and is guarded against loss or access thanks to the system's combination of encryption techniques, access controls, authentication systems, and backup/disaster recovery procedures. Additionally, Privacy guidelines and regulations such as GDPR (General Data Protection Regulation) are designed to protect the personal information of individuals and ensure that it is collected, processed, and stored in a secure and lawful manner. To comply with GDPR and other privacy regulations, the system implements several measures to protect users' personal information.

IV. SCALABILITY AND MAINTENANCE

The MIT-GPT project is designed to be , allowing for future expansion and integration with other college systems or databases. The system architecture is modular, making it easy to add or update functionalities as needed. Maintenance tasks, such as model retraining, dataset updates, and bug , are performed periodically to ensure the system's performance and accuracy. Automated monitoring and logging are also implemented to detect and address any potential issues in real time. The system's performance and accuracy must be maintained over time through tasks. Automated monitoring and logging systems can assist in real-time problem detection and resolution, lowering the likelihood of system failure or downtime.

A. Future Enhancements

The MIT-GPT project has the potential for future enhancements and expansions. Some of the possible future directions include: (a)Integration with other college systems, such as the student information system, library system, or learning management system, to provide a comprehensive and seamless experience for users. (b)Incorporation of machine learning techniques, such as reinforcement learning or active learning, to further improve the model's accuracy and relevance. [5] (c)Expanding to support multi-lingual queries allows users to input queries in different languages and receive responses in their preferred language. (d)Incorporating natural language processing techniques, such as sentiment analysis or entity recognition, enhances the assistant's understanding of user queries and provides more responses.

CONCLUSION

The MIT-GPT project is an innovative college query assistant that combines the OpenAI API and LangChain to deliver precise and pertinent answers to questions about higher education. The architecture of this system is built to respond to user queries quickly and accurately while protecting the security of user data. The system is based on a deep learning model that has been trained on a sizable dataset of data pertaining to colleges, enabling it to offer thorough and precise solutions to a variety of inquiries. The system draws on both openly accessible data from places like government databases and college websites, as well as private information that the development team has collected and selected. The system's model customization makes sure that it can still respond to users' questions in a pertinent and correct manner even when presented with fresh or unfamiliar facts. The development of the system must include performance evaluation in order to make sure that the system is achieving its accuracy and performance objectives. As it reveals user preferences and potential areas for system improvement, user feedback is a crucial component of system development. Users may quickly and easily discover the information they need because to the system's user interface's intuitive and simple design. To guarantee that user data is kept secure and confidential, security measures are put in place. These methods include encryption

techniques and access controls. The system's ability to be quickly scaled to meet the demands of an expanding user base is another important factor. To guarantee that the system continues to function effectively and efficiently over time, maintenance operations including model retraining and bug repairs are carried out on a regular basis.

The MIT-GPT assistant has the potential to significantly improve the college experience for students, teachers, and staff by providing quick and accurate information. To satisfy the changing needs of users, the system may undergo further development in the form of new data sources, enhanced natural language processing capabilities, and increased functionality.

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